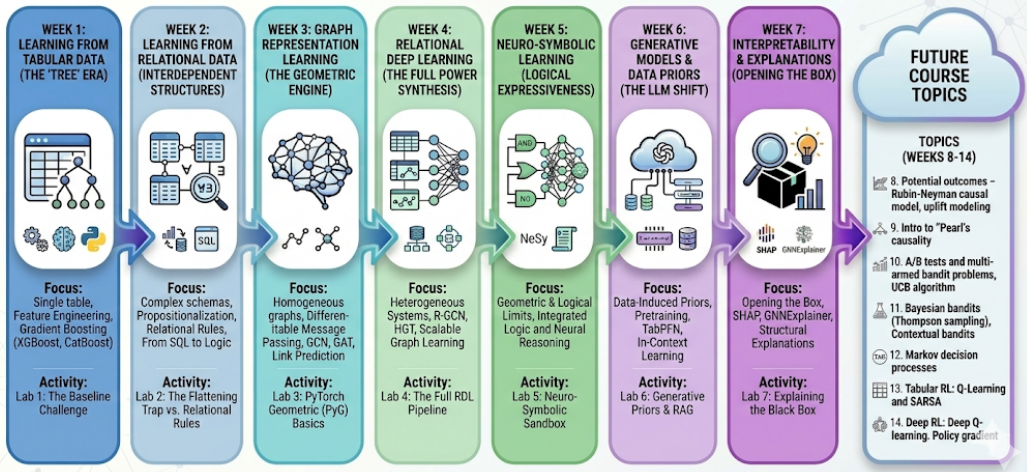


LEARNING FROM STRUCTURED DATA: A 7-WEEK JOURNEY

Narrative Arc: Increasing Structure → Increasing Inductive Bias → Increasing Reasoning Power



Course Map of Week 7: Opening the Black Box

- 1 Week 7 Module 1: Introduction & XAI Motivation
- 2 Week 7 Module 2: Post-Hoc Model-Agnostic XAI (LIME, SHAP)
- 3 Week 7 Module 3: Post-Hoc Model-Specific (Structural) XAI
- 4 Week 7 Module 4: Inherently Interpretable Models (Glass Box)
- 5 Week 7 Module 5: Concept-Based (Mechanistic) Interpretability
- 6 Week 7 Module 6: Course Conclusion

Week 7: Opening the Black Box

The Course Narrative:

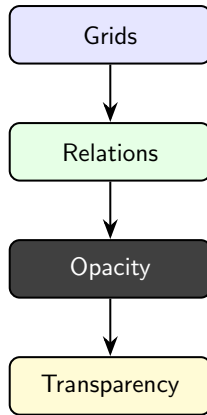
- **W1-W2:** Tabular & Relational (Structure)
- **W3-W5:** GNNs & NeSy (Topology)
- **W6:** Foundation Models (Data priors)

The Cost of Complexity:

- Complex reasoning creates high-dim. manifolds
 - Decoding them requires complex opaque models
- Millions of parameters → "Black Box" problem.

Our Final Goal:

- Shift from *what* is predicted to *why* and *how*.
- Decode the structures we built.



The Demand for Transparency

Definition

Interpretability is the degree to which a human can understand the cause of a decision.

Trust & Verification:

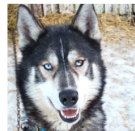
- High accuracy $\neq$ Correct reasoning (Clever Hans effect).
- High-stakes domains: Medicine, Criminal Justice ...

Model Debugging:

- Identifying spurious correlations, data leakage, and bias.

Compliance & Ethics:

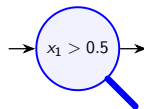
- AI Act's "Right to Explanation", GDPR compliance.
- What ethical values does the model utilize?



(a) Husky classified as wolf



(b) Explanation



The Evaluation Crisis: Fidelity vs. Plausibility

The Lack of Ground Truth:

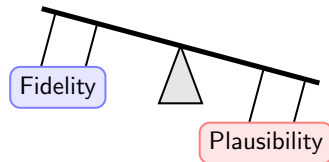
How do we evaluate an explanation?

Fidelity (Faithfulness):

- Does the explanation accurately reflect the model's internal computation?

Plausibility:

- Does the explanation look reasonable and convincing to a *human*?



The Paradox

Optimizing for plausibility can destroy fidelity.

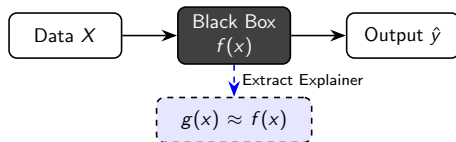
Question: *If AI denies a loan and generates a perfectly logical reason, but it's not the actual reason it used—is that an explanation, or deception?*

The Two Paths of Explainable AI (XAI)

1. Post-Hoc Explanations

(Surrogate/Opening the Black Box)

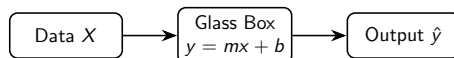
- **Approach:** Train a complex model $f(x)$, approximate it after-the-fact with a simpler explainer $g(x)$.
- **Models:** XGBoost, MLP, GNN, etc.
- **Examples:** LIME, SHAP, GNNExplainer.
- **Flaw:** $g(x)$ is always just an approximation of $f(x)$, otherwise we would have a glass box.



2. Inherently Interpretable

(Glass-Box Approach)

- **Approach:** Constrain hypothesis space F during design or training $\rightarrow$ model $f(x)$ is naturally transparent.
- **Examples:** Sparse Trees, Rule Lists, Linear Models.
- **Philosophy:** The only reliable explanations (e.g. for high-stakes decisions) are the models themselves



Deconstructing the Tabular Box

Recap: Week 1 (Tabular Models)

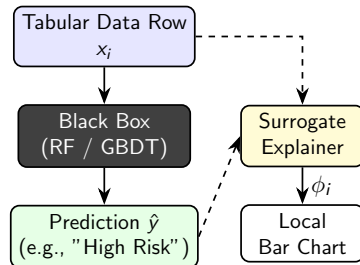
- Powerful bagging&boosting models (RF, XGBoost).
- **Problem:** High accuracy, but opaque internals.

Global vs. Local Feature Importance

- *Global:* Which features matter overall? (e.g., XGBoost Gain). Hides subgroup heterogeneity.
- *Local:* Why did the model predict A for patient B ?

The Post-Hoc Surrogate Approach

- Treat trained model as frozen Black Box (oracle).
- Probe it with modified inputs to understand local topography.



LIME: Local Surrogate Intuition

The Core Idea (Ribeiro et al., 2016)

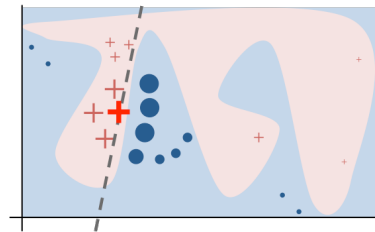
- Complex global decision boundary is hard to explain.
- But *locally*, any smooth curve can be approximated by straight line.

The LIME Algorithm

- 1 Take the instance of interest x .
- 2 Sample synthetic neighborhood by perturbing features.
- 3 Get black-box predictions for these synthetic points.

Local Weighting

- 4 Weight synthetic samples by geometric distance to x .
- 5 Fit a sparse linear model to these weighted samples.



The Mathematics of LIME

The Optimization Objective

LIME finds an explanation g (from a family of simple models G) by optimizing:

$$\xi(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

Deconstructing the Equation:

- $f(x)$: The original opaque black-box model (e.g., XGBoost).
- $g(z')$: The interpretable surrogate model (e.g., Linear Regression).
- $\mathcal{L}$: The fidelity loss function (e.g., Mean Squared Error).

Locality and Sparsity:

- $\pi_x(z)$: The proximity kernel (exponential distance weight). Forces *local fidelity*.
- $\Omega(g)$: The complexity penalty (e.g., L1 Lasso regularization). Forces *sparsity*.

SHAP: The Game-Theoretic Foundation

The Limitation of LIME:

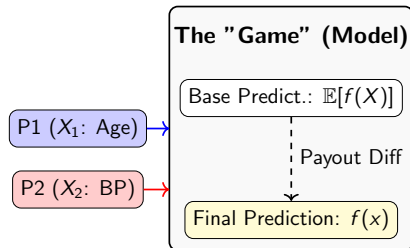
- LIME is heuristic; lacks theoretical guarantees for fairness in feature attribution.

Cooperative Game Theory (L. Shapley, 1953)

- **Game:** Prediction for a data instance.
- **Players:** Features (variables) of the instance.
- **Payout:** Difference between actual prediction and global average prediction.

The Shapley Value:

- *Fair* distribution of prediction payout among features based on their contributions.



Computing the Shapley Value

The Shapley Equation

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x) - f_S(x)]$$

Deconstructing the Math:

- ϕ_i : The Shapley value (fair contribution) for feature i .
- F : The set of all features. S : A specific coalition (subset) of features without i .

Marginal Contribution:

- $[f_{S \cup \{i\}}(x) - f_S(x)]$: The change in prediction when feature i joins coalition S .

Combinatorial Weight:

- $\frac{|S|!(|F| - |S| - 1)!}{|F|!}$: Weighs marginal contribution by how often the coalition occurs.

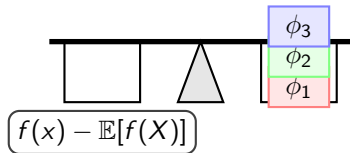
SHAP's Unifying Axioms

Additive Feature Attribution Methods

SHAP represents the explanation as a linear model of simplified coalitions:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i$$

where $z'_i \in \{0, 1\}$ indicates if feature i is present.



Why is SHAP the only correct answer?

Game theory proves Shapley values are the *unique* solution satisfying three axioms:

- 1 **Efficiency (Local Accuracy):** Sum of all feature contributions $\sum \phi_i$ must be equal to difference between specific prediction and baseline average.
- 2 **Missingness:** If a feature is missing $z'_i = 0$, it is assigned a 0 contribution.
- 3 **Consistency:** If a model change increases feature i 's marginal contrib., SHAP value ϕ_i must not decrease.

SHAP in Practice (TreeSHAP)

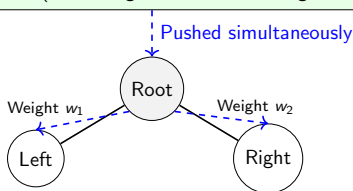
The Computational Crisis:

- Evaluating all subsets requires $\mathcal{O}(2^M)$ model calls (NP-hard).

Approximations:

- *KernelSHAP*: LIME-style weighted regression to approximate Shapley values.
- *TreeSHAP*: Exact, polynomial-time algorithm for tree ensembles (e.g., XGBoost).
 - Pushes all possible subsets S down the decision tree simultaneously.
 - Complexity drops from $\mathcal{O}(TL2^M)$ to $\mathcal{O}(TLD^2)$ (where D is max tree depth).

Data Point x (evaluating subset with missing feature)



The Fatal Flaw: The I.I.D. Illusion

The Assumption of Independence

- To evaluate a feature (e.g., X_j), SHAP/LIME sample from its marginal distribution $\mathbb{E}[f(X)]$.
- Assumes X_j is independent of other features X_{-j} .

The Reality of Structured Data

- Tabular/relational data are highly correlated.
- Sampling marginals creates impossible, OOD "Frankenstein" instances.



Engagement Question:

If we train model on housing data and mask 'area' feature by randomly sampling from dataset with 'num_bedrooms=5' fixed, can we create a 5-bedroom house with 100ft²?

Complex structures require Structural Explanations (Week 3/4 on GNNs).

The Transition to Topology

The Limit of LIME/SHAP:

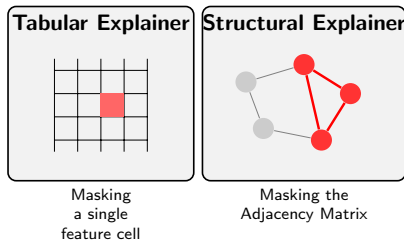
- Assumes features are independent (I.I.D.).
- Masking features via marginal sampling destroys structural context.

The Reality of Week 3 & 4:

- In RDL (Week 4), topology *is* the feature.
- Predictions rely on paths, cycles, and neighborhoods.

New Paradigm: Structural Explanations

- Instead of asking "Which *feature* mattered?", we ask "Which *edges/subgraph* caused the prediction?"



The Explanation Search Space (Receptive Field)

Bounding the Search Space:

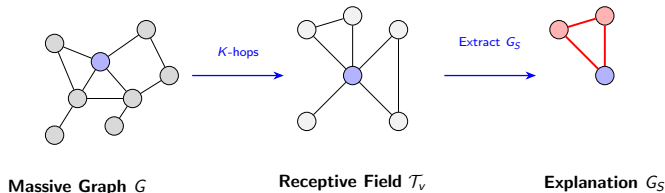
- A graph $G(V, E)$ can be massive (millions of nodes). Where do we look?

Recall Week 3 (The MPNN Blueprint):

- A K -layer GNN aggregates messages from a K -hop neighborhood.
- The prediction for node v strictly depends *only* on its computation tree $\mathcal{T}_v$.

The Structural Constraint:

- Any valid explanation subgraph G_S must be a subset of the K -hop computation tree: $G_S \subset \mathcal{T}_v$.



GNNExplainer: The Objective

The Intuition (Ying et al., 2019):

- Find small subgraph G_S that retains critical information about model's original prediction.

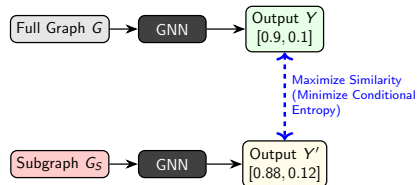
Mutual Information (MI) Maximization:

- We want to maximize the MI between the prediction Y and the subgraph G_S :

$$\max_{G_S} MI(Y, (G_S, X_S))$$

$$MI = H(Y) - H(Y|G = G_S, X = X_S)$$

- $H(Y)$: Entropy of original prediction (Fixed).
- $H(Y|G_S)$: Conditional entropy. We minimize this by forcing subgraph's prediction to be equal to full-graph's prediction.



Continuous Relaxation of Discrete Edges

The Optimization Bottleneck:

- Searching over all possible discrete subgraphs (edge is 0 or 1) is NP-hard ($\mathcal{O}(2^{|E|})$).

The Masking Solution:

- Instead of hard subset selection, learn a continuous fractional mask $M \in \mathbb{R}^{|E|}$.
- Masked adjacency matrix becomes: $A_S = A \odot \sigma(M)$.

$$\begin{array}{c}
 \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 0.1 & 0 \\ 0.1 & 1 & 0.9 \\ 0 & 0.9 & 1 \end{bmatrix} \\
 \text{Raw Adj (A)} \quad = \quad \text{Mask } \sigma(M) \\
 \\
 \begin{bmatrix} 1 & 0.1 & 0 \\ 0.1 & 1 & 0.9 \\ 0 & 0.9 & 1 \end{bmatrix} \\
 \text{Weighted Graph } A_S
 \end{array}$$

Backprop (Gradient Descent)

Differentiable Search: $\sigma(M)$ is continuous, we can use GD to optimize the mask!

$$\mathcal{L} = - \sum_{c=1}^C \mathbb{I}[y = c] \log P(Y = c | A_S) + \lambda_1 \|M\|_1 + \lambda_2 H(M)$$

- Minimize prediction loss + Size constraint (L1) + Binarization (Entropy).

GNNExplainer in Action (Visualizing the Mask)

What does the output look like?

- The optimized mask M acts as a set of edge importance weights.

Thresholding for Readability:

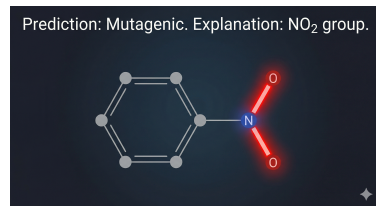
- Edges with $m_{ij} > \tau$ (e.g., 0.5) are kept; others are discarded.

Fidelity Check (Lab 7 Preview):

- If we delete high-weight edges from the original input graph, does GNN's prediction flip?

Applications:

- *Chemistry*: Identify toxic groups (mutagenicity).
- *Finance*: Highlight transaction loops (fraud).



The Transductive Bottleneck (Scalability)

The Limitations of GNNExplainer:

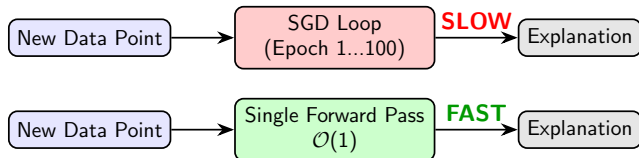
- It is an **Instance-Level (Transductive)** optimization.
- Mask M is learned via Gradient Descent *from scratch* for every explained node.

The Computational Cost:

- Each of 1,000 fraud alerts requires its GD optimization (1,000s epochs each).

The Inductive Mismatch:

- In Week 3, we engineered GNNs to be *Inductive* (generalizable to new nodes).
- Our explainer is stuck being Transductive. We need a global, inductive explainer.



PGExplainer (Parameterized Global Explainer)

The Core Idea (Luo et al., 2020):

- Instead of optimizing a mask M directly, train a Neural Network to *predict* M .

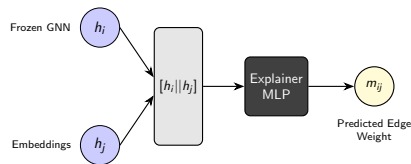
The Parameterized Explainer Network:

- Take embeddings of trained GNN for two connected nodes (h_i, h_j).
- Conca. and pass through an MLP explainer:

$$m_{ij} = \text{MLP}([h_i || h_j])$$

Global Inductive Learning:

- MLP is trained across multiple graphs to maximize Mutual Information globally.
- **Explaining new node requires no SGD. Just one $\mathcal{O}(1)$ fwd pass thru MLP!**



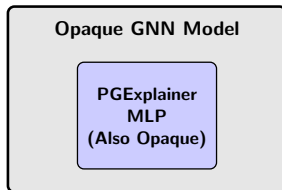
Explaining a Black Box with a Black Box?

Engagement Question:

PGExplainer trains an MLP... haven't we built just another Black Box?

The Reality of Post-Hoc Explainers:

- Yes. PGExplainer is a black-box meta-model, which suffers from the "Faithfulness-Plausibility Gap" (Module 1).
- it might learn a heuristic shortcut that doesn't reflect the GNN's true logic.



The "Glass Box" Philosophy

The Post-Hoc Problem (Rudin, 2019):

- "Stop explaining black box machine learning models for high stakes decisions".
- Explanations of black boxes are imperfect approx.

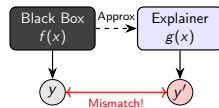
Double Trouble:

- Deploying explainer relies on 2 models instead of 1.
- Inevitable disagreement on edge cases → fidelity.

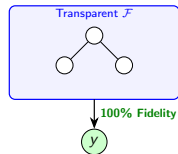
The Alternative: Inherently Interpretable Models

- Design the hypothesis space $\mathcal{F}$ so the model provides its own computational pathway.
- **Goal:** Perfect mathematical fidelity by design.

Post-Hoc ("Double Trouble")



The "Glass Box"



The Rashomon Effect

The Myth of the Tradeoff:

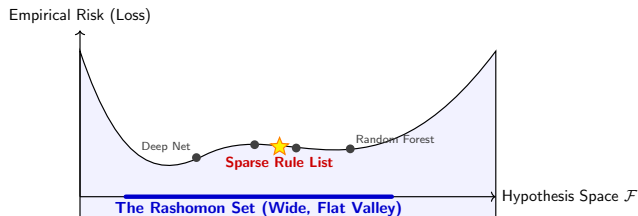
- It is a fallacy that interpretability *always* costs accuracy.

The Rashomon Set:

- On noisy, finite datasets, a large set of models has near-optimal empirical risk.
- This set of equally good models is called the "Rashomon Set".

Finding the Glass Box:

- Rashomon set is large $\rightarrow$ likely contains at least one sparse, interpretable model.
- We just need better optimization algorithms to find it!



Optimal Rule Lists & Scoring Systems

Certiably Optimal Rule ListS (CORELS)

- Uses branch-and-bound optimization to find the *global* optimal IF-THEN list.
- Guarantees maximum accuracy for a given sparsity penalty.

Supersparse Linear Integer Models (RiskSLIM)

- Linear classifier restricted to small int. weights.
- Allows calculating risk via simple addition.

Applications

- *CORELS*: Predicting recidivism without black-box biases.
- *RiskSLIM*: ICU Seizure prediction scores.

CORELS Model:

```
IF age = 18-20 AND sex = male  
  THEN predict High Risk  
ELSE IF prior_crimes > 3  
  THEN predict High Risk  
ELSE predict Low Risk
```

RiskSLIM Scorecard:

Condition	Points
Age > 60	+1
High Blood Pressure	+2
Prior Seizure	+2
SCORE	...
If SCORE > 2 → Predict Seizure.	

The imodels Ecosystem

Bridging Theory to Practice:

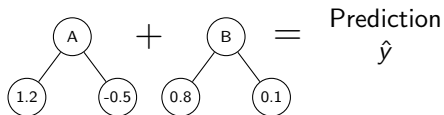
- Historically, optimal sparse models were fragmented and computationally inefficient.
- **imodels**: A unified, scikit-learn compatible Python library for interpretable modeling.

Key Algorithms Available:

- **RuleFit**: Extracts rules from a Random Forest and fits a sparse L1 linear model.
- **FIGS (Fast Interpretable Greedy-Tree Sums)**: Generalizes CART to grow a sum of very small trees simultaneously, capturing additive structures perfectly.

Lab 7 Preview:

```
from imodels import FIGSClassifier
model = FIGSClassifier(max_rules=10)
model.fit(X_train, y_train)
```



Model Distillation (Global Surrogates)

The Strategy: Teacher → Student

- Train a complex, high-dimensional Black Box (the Teacher).
- Predict on a large pool of unlabeled data (Soft Labels).
- Train a transparent Rule List (the Student) on the Teacher's soft labels.

Why it works:

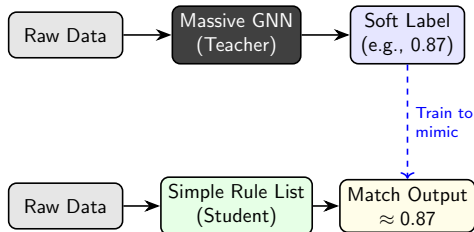
- Student distills complex feature engineering discovered by GNN into logic.

The Fidelity Guarantee:

- This *Global Surrogate* explains entire model's behavior natively.

Engagement Question:

If the Student Rule List is so accurate, why not to train directly on original raw data and skip the Teacher entirely?



Proof Traces (The NeSy Advantage)

Revisiting Week 5:

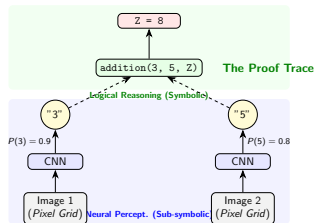
- Neuro-Symbolic (NeSy) models bypass post-hoc explainers entirely.

DeepProbLog & Inherent Transparency:

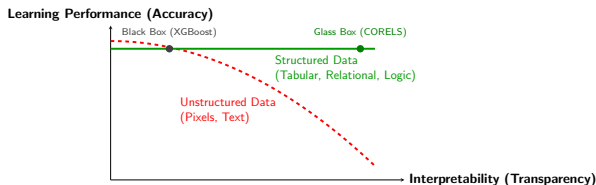
- Neural networks handle sub-symbolic perception (e.g., reading an image).
- Probabilistic Logic Programming handles reasoning.

The Proof Trace:

- Logical execution path (proof tree) connecting perception to conclusion *is* the explanation – it is the actual mechanism, guaranteeing 100% fidelity.



The Accuracy-Interpretability Tradeoff Re-evaluated



The Consensus on Structured Data:

- For Tabular, Relational, and Logical data, the tradeoff curve is largely flat.
- With modern optimiz. (e.g., `imodels`), no sacrifice of accuracy for transparency.

The Transition to Unstructured Data:

- For raw sequences (text) and grids (pixels), finding inherent Glass Box is difficult.
- **Next Module:** How do we interpret the massive generative language models (LLMs) from Week 6 when inherent logic isn't enough?

Explaining the Generative LLM

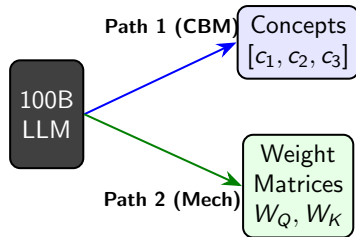
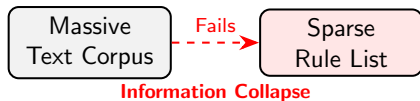
The Limit of the Glass Box:

- Week 5/6: Transition to large Transformers trained on unstructured text/pixels.
- Cannot enforce strict sparsity on a 100B parameter LLM without huge accuracy loss.

Two New Frontiers for Unstructured Data:

- **1. Concept Bottleneck Models (CBMs):**
Force the NN to predict human-readable *concepts* before the final output.
- **2. Mechanistic Interpretability:**
Reverse-engineer the matrix multiplications directly.

Goal: Decode generative priors from Week 6.



Concept Bottleneck Models (CBMs)

The Intuition:

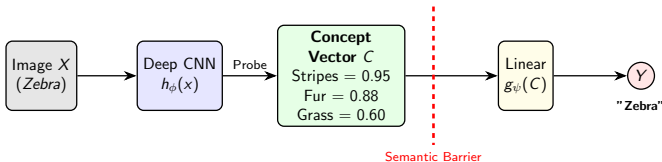
- Standard DNN: $X \rightarrow Y$ (Opaque).
- Concept Bottleneck: $X \rightarrow C \rightarrow Y$ (Interpretable intermediate steps).

The Architecture (Koh et al., 2020):

- **Step 1 (Probing):** Train $h_\phi(x)$ to predict a vector of known concepts C (e.g., "bone spurs", "fur", "stripes").
- **Step 2 (Prediction):** Train a simple, sparse classifier $g_\psi(C)$ to predict the final target Y solely from the concepts.

The Advantage:

- The final decision relies *only* on human-understandable semantic variables, not abstract latent geometry.



Human-AI Collaboration via Intervenability

The Power of Intervenability:

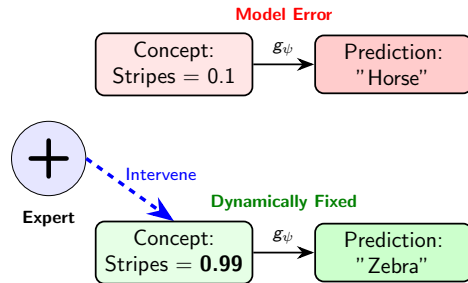
- Because bottleneck C is semantic, humans can interact directly with model's intermediate reasoning.

Test-Time Intervention:

- If the model incorrectly predicts c_i ("stripes" = false), a human expert can manually correct the concept to c'_i ("stripes" = true).
- The downstream classifier $g_\psi(c')$ instantly updates the final prediction y' .

Stochastic CBMs:

- Recent advances map concepts to probability distributions, allowing us to quantify uncertainty on specific concepts.



The Catch

Requires massive, dense annotation of concepts C for every training instance.

Mechanistic Interpretability (The Ultimate Frontier)

When CBMs are impossible:

- For internet-scale LLMs (Week 6), we cannot annotate concepts for trillions of tokens.

The Mechanistic Philosophy (Olah / Nanda):

- Treat a trained Neural Network not as a black box, but as a compiled binary program.
- **Goal:** Reverse-engineer the raw weight matrices into human-readable algorithms (Circuits).

The Agenda:

- Find *Features* (neurons/directions) and trace the *Circuits* (computational graphs connecting them).

The Residual Stream: The Communication Bus

Residual Stream View (Elhage et al., 2021):

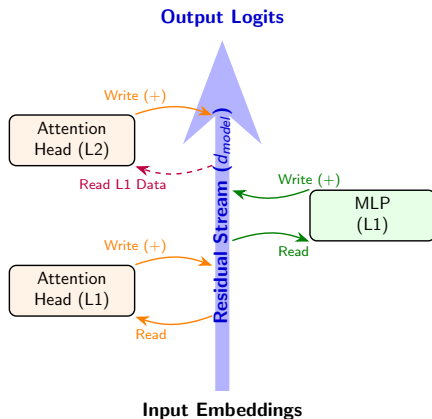
- The central d_{model} -dimensional vector passing through all layers.
- Functions as a *Shared Memory/Bandwidth*.

Read/Write Operations:

- Attention heads and MLPs *read* from specific subspaces of the stream.
- They compute results, and *add* (write) them back into the stream.

The Independence Postulate:

- Information persists in the stream unless actively overwritten. Layers communicate with each other linearly by writing vectors that future layers will read.



Deconstructing Attention: QK and OV Circuits

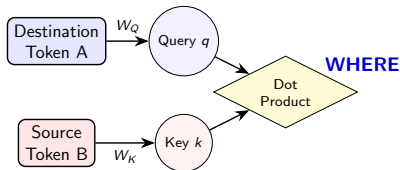
Splitting the Attention Head (Anthropic Framework)

An Attention Head computes two completely independent, linear operations.

1. The Query-Key (QK) Circuit

$$W_{QK} = W_Q^T W_K$$

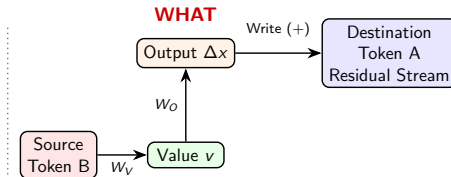
- Computes the Attention Score.
- **Function:** *where* to move information *from* and *to* (Source → Destination).



2. The Output-Value (OV) Circuit

$$W_{OV} = W_O W_V$$

- Computes impact on the residual stream.
- **Function:** *what* information is extracted from Source and written to Destination.



Skip-Trigrams & The 1-Layer Limit

The Power of 1-Layer Transformers

- By multiplying the QK and OV circuits with the Embedding/Unembedding matrices, we find **Skip-Trigrams**.

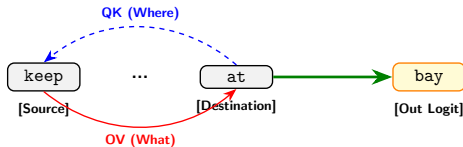
The Format: [source] ... [destination]
→ [out]

- QK Circuit* connects [destination] to [source].
- OV Circuit* reads [source] and increases logits for [out].
- Examples:* keep ... at → bay, or def ... (→ self.

Engagement Question:

If an Attention Head only moves information from a Source to a Destination, can a 1-Layer Transformer perform In-Context Learning (e.g., repeating a novel pattern "X Y ... X → Y")?

Answer: No. It requires **Composition** (multiple layers).



The Phenomenon of In-Context Learning

Recap Week 6 (In-Context Learning):

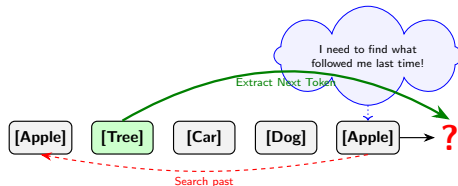
- LLMs can recognize and apply novel patterns at inference time *without* any weight updates.

The Abstract Pattern:

- Prompt: [A] [B] ... (random text) ... [A] $\rightarrow$?
- The model dynamically learns to predict [B].

The Mechanistic Question:

- How does the final token [A] know to predict [B] if the QK/OV circuits only map fixed tokens?
- It must somehow *find* the previous [A], and *extract* the token that came immediately after it.



The Induction Head (The 2-Layer Circuit)

The Induction Circuit (Olsson et al., 2022)

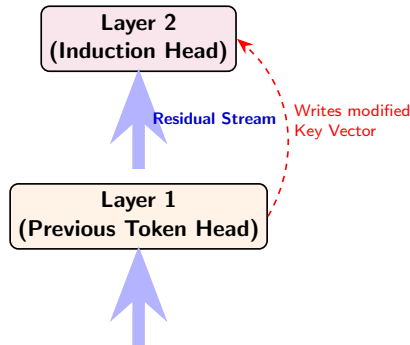
- A mechanism specifically responsible for the majority of In-Context Learning.

The Setup:

- Requires at least 2 Attention Layers to function via **Composition**.

The Two Actors:

- 1 **The Previous Token Head (Layer 1):** Moves information from token $t - 1$ to token t .
- 2 **The Induction Head (Layer 2):** Uses the information from Layer 1 to find the correct token and copy it to the output.



Circuit Step 1: The Previous Token Head

Action: Marking the Predecessor

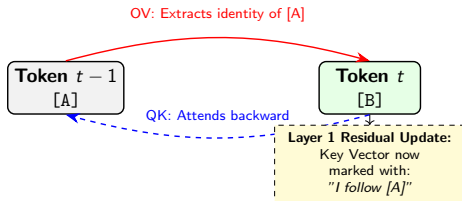
- In Layer 1, the **Previous Token Head** attends strictly to the token immediately behind it ($t - 1$).

The OV Extraction:

- It extracts the identity of token $t - 1$ (e.g., [A]) and writes it into the residual stream of token t (e.g., [B]).

The Result (The "Mark"):

- The Key vector of token [B] now contains a new, localized feature: *"I am the token that comes immediately after [A]"*.



Circuit Step 2: K-Composition

Action: The Induction Match

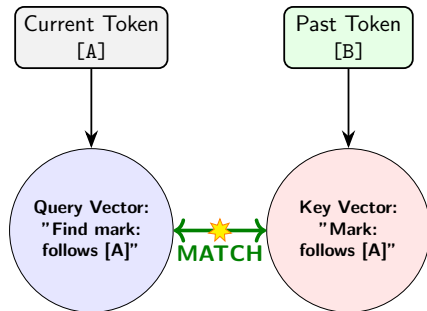
- We fast-forward to the end of the prompt, where the model sees the second [A].

K-Composition (Layer 2):

- The **Induction Head** (Layer 2) generates a Query for the concept [A].
- It searches all past Key vectors for the feature: *"Who has the mark indicating they follow [A]?"*.

The Match:

- It perfectly matches with the Key vector of token [B] (which was modified by Layer 1).
- **Result:** The QK circuit successfully routes attention to the correct token [B].



Circuit Step 3: OV Copy & Conclusion

Action: Copying to the Output

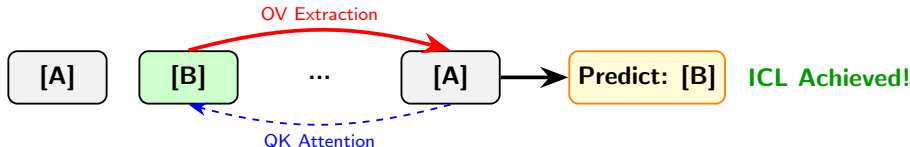
- Now that the Induction Head is attending to [B], its **OV Circuit** activates.

The Write Operation:

- It extracts the semantic identity of [B], routes it to the final token [A], and writes it into the final logits.
- Prediction:** The model outputs [B].

Conclusion: The Glass Box of Generative AI

- By tracking vectors through QK/OV circuits, we proved *exactly* how LLMs learn in-context.
- We transformed a black-box Transformer into an auditable, deterministic algorithm.



The Evolution of Inductive Bias

Week 1: Flat Grids ("Dark Matter")

- *Structure*: Independent, heterogeneous rows.
- *Prior*: Axis-aligned splits (Trees, XGBoost).

Weeks 2–4: Topology & Geometry

- *Structure*: Interdependent entities, relationships.
- *Prior*: Differentiable message passing (GNNs, RDL).

Week 5: Neuro-Symbolic Logic

- *Structure*: Formal reasoning, strict constraints.
- *Prior*: Logical operators as structural regularizers.

Week 6: Generative Models (LLMs)

- *Structure*: Unstructured sequences.
- *Prior*: Internet-scale data priors (Transformers).

The Final Lesson: Aligning Model and Explanation

The Fallacy of the Universal Model:

- There is no single "best" algorithm.
- Performance req. matching model's *Inductive Bias* to data's true structure.

XAI Mandate (Structured Data):

- For high-stakes tabular/logical tasks, use **Inherently Interpr. Models**.
- Accuracy-interpretability tradeoff is a myth here.

XAI Mandate (Unstructured Data):

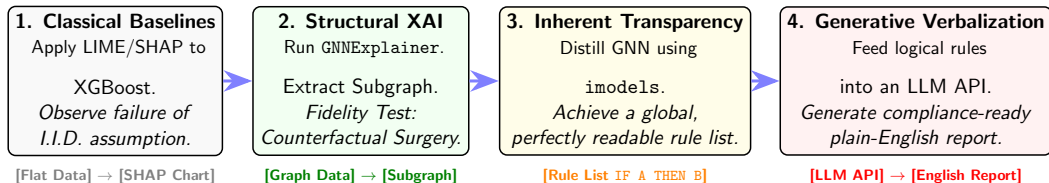
- For LLMs, use **Mechanistic Interpr.** to reverse-engineer causal circuits.

Engagement Question:

If a global bank needs to predict credit defaults using 100 million rows of highly structured financial history, should they deploy a fine-tuned LLM or a Sparse Rule List distilled from an XGBoost model?

	Structured Data	Unstructured Data
Best XAI Best Model	XGBoost / GNN (Strong Priors)	Transformers / LLMs (Data Priors)
	Glass Box (Optimal Rule Lists)	Mechanistic Interpretability

Lab 7: The Ultimate XAI Pipeline



See you in the final Lab!